

BIS603/603: Exercise 4 ERD

September 6, 2025

According to the below scenario,

1. identify all relationships of each entity within your database design using the Crow's Foot notation. Include all data elements (attribute), entities, relationships (such as 1:M, 1:1, M:N), cardinalities, and participatory for the gallery store database in your diagram.
2. Give 5 information that you want from this database. Give clearly explanation.

Although you always wanted to be an artist, you ended up being an expert on databases because you love to cook data and you somehow confused *database* with *data baste*. Your old love is still there, however, so you set up a database company, ArtBase, that builds a product for art galleries. The core of this product is a database with a schema that captures all the information that galleries need to maintain.

Galleries keep information about artists, their names (which are unique), birthplaces, age, and style of art. For each piece of artwork, the artist, the year it was made, its unique title, its type of art (e.g., painting, lithograph, sculpture, photograph), and its price must be stored. Pieces of artwork are also classified into groups of various kinds, for example, portraits, still lifes, works by Picasso, or works of the 19th century; a given piece may belong to more than one group.

Each group is identified by a name (like those just given) that describes the group. Moreover, galleries keep information about customers. For each customer, galleries keep that person's unique name, address, total amount of dollars spent in the gallery (very important!), and the artists and groups of art that the customer tends to like. Finally, the gallery gives the special session for the customer who apply for membership such as some promotion (discount, coupon). Each customer must register to be the member of gallery before buying the piece of art. The customer who do not enroll for the member can do only site visit via the gallery website.